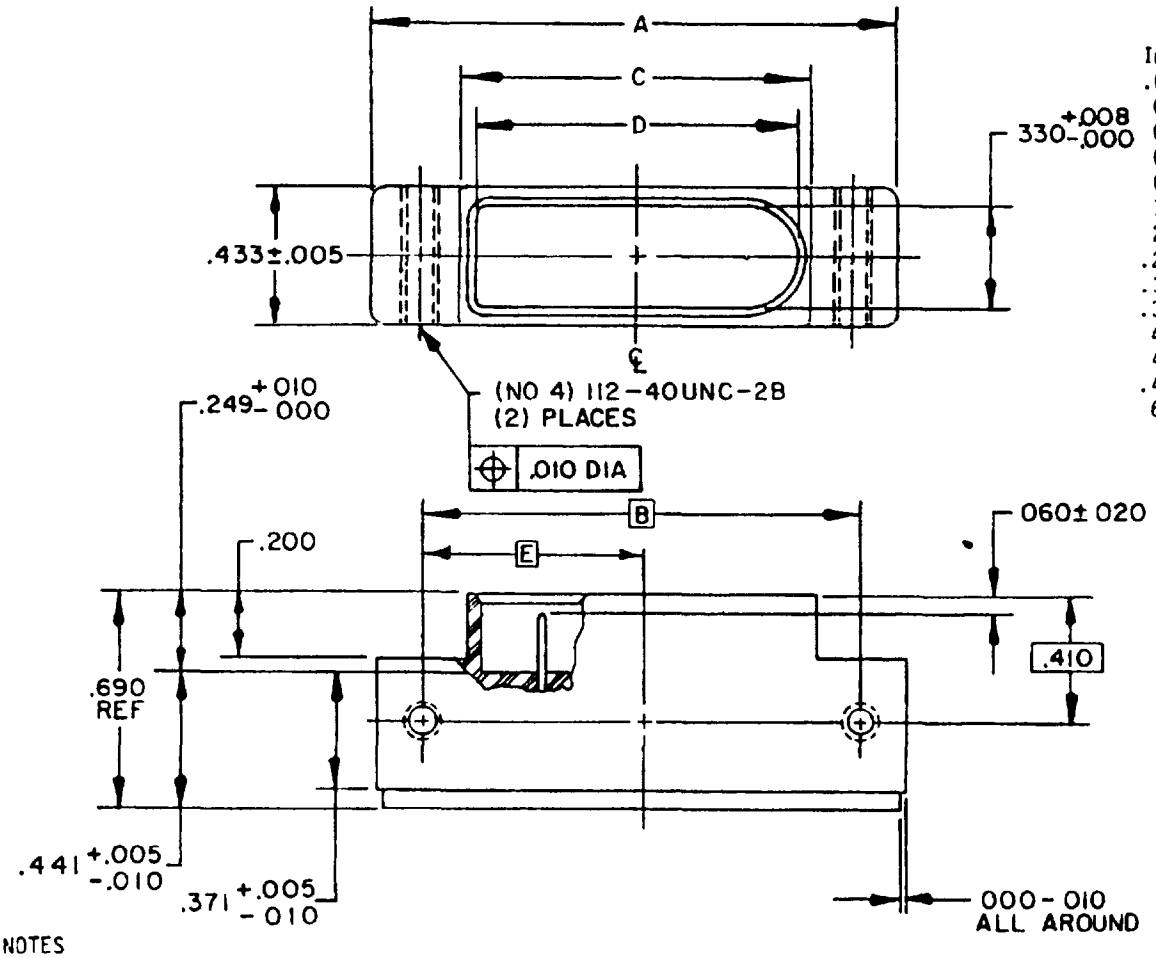
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PIN PLUG, ELECTRICAL POLARIZED (.090 SPACING), REMOVABLE CRIMP CONTACTS

Ⓒ This specification inactive for new design after 1 October 1986

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-55302.



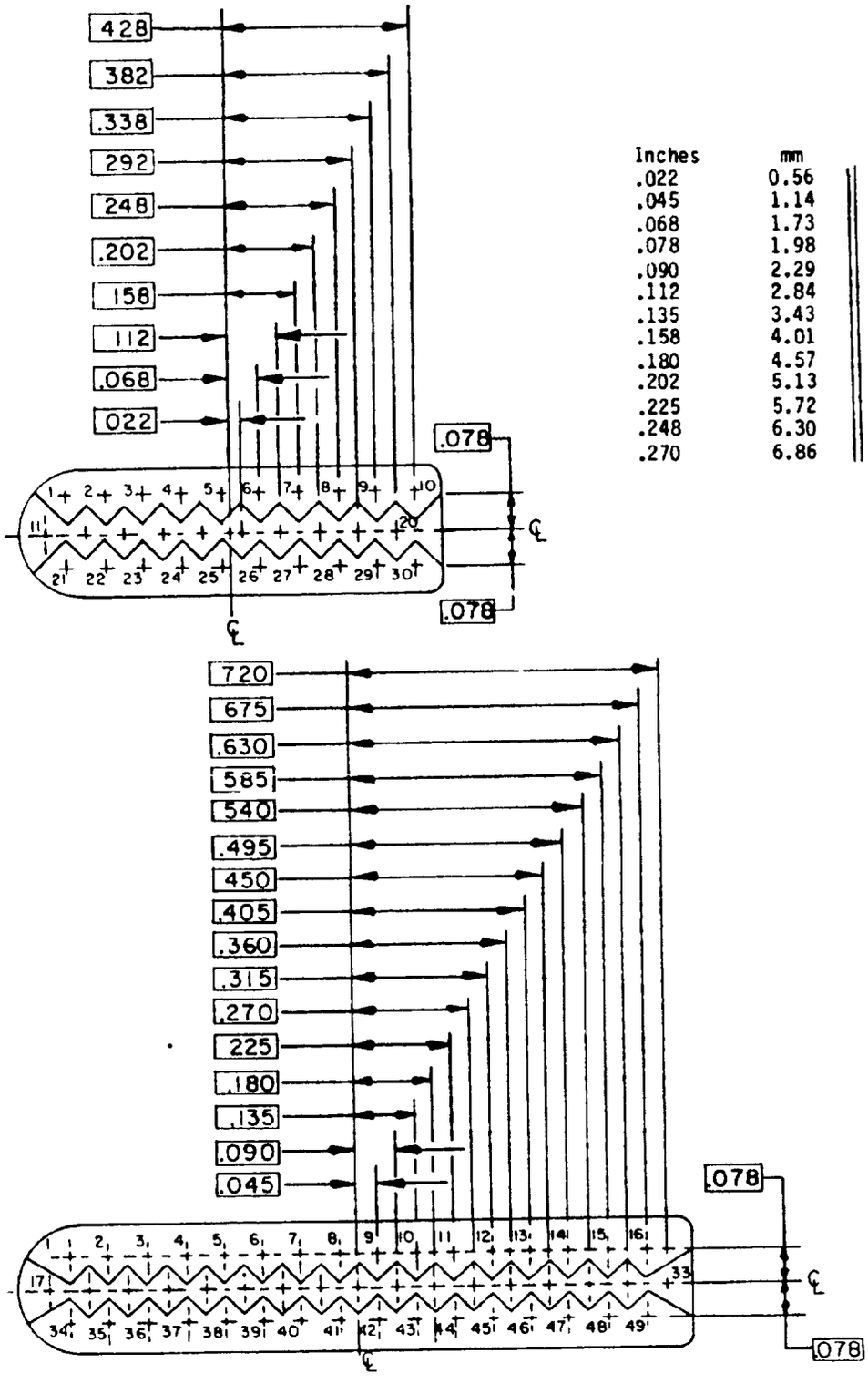
Inches	mm
.005	0.13
008	0 20
010	0 25
020	0 51
060	1 52
112	2.84
200	5.08
.249	6.32
.330	8 38
.371	9.42
410	10 41
433	11.00
.441	11.20
690	17.53

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Nominal spacing between any two adjacent pins contacts shall be .090 ±.003 (2.29 ±0.08 mm).
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. For dimensions A through E, see table I
6. A quantity of MS27493-22D pin crimp contacts consisting of the normal complement plus one spare contact for connector arrangements having 26 contacts or less and two spares for arrangements having over 26 contacts shall be supplied with each connector.

FIGURE 1. Connector, plug, .090 (2.29 mm), contact spacing, removable crimp pin.

Ⓒ denotes changes

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Inches	mm	Inches	mm
.022	0.56	.292	7.42
.045	1.14	.315	8.00
.068	1.73	.338	8.59
.078	1.98	.360	9.14
.090	2.29	.382	9.70
.112	2.84	.405	10.29
.135	3.43	.428	10.87
.158	4.01	.450	11.43
.180	4.57	.495	12.57
.202	5.13	.540	13.72
.225	5.72	.585	14.86
.248	6.30	.630	16.00
.270	6.86	.675	17.15
		.720	18.29

NOTES:
1. Contacts 1, 20, and 21 (Top of Page) are identified on the front face of the connector and all contacts identified on the rear face per the arrangement.
2. Contacts 1, 16, and 49 (Bottom of Page) are identified on front face of the connector and all contacts are identified on the rear face per the arrangement.

FIGURE 2. Insert arrangement for .090 (2.29 mm) contact spacing connectors (Male engaging face).

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302, except:

Body - reinforced epoxy resin or equivalent.

Bushing - monel, QQ-N-281, class B.

Plating: In accordance with MS27493-22D.

Contact identification: Contact locations are identified numerically as shown on figure 2.

Mating and unmating: The maximum mating force, in pounds, shall be 0.5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: With a test current of 3 A dc, the average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 V rms, 60 Hz, ac.

High altitude: 275 V rms, 60 Hz, ac.

Contact current rating: 3.0 amperes.

Insert arrangement: See figure 2.

Mating connectors: See MIL-C-55302/68.

Contact insertion and removal tool: See MIL-C-55302/79.

Quality conformance inspection:

Group A inspection: Delete mating and unmating.

Group B inspection: Add mating and unmating.

Part number: M55302/69-(dash number from table I).

TABLE I. Dash number and dimensions.

Dash number	Dimensions 1/				
	+.010 A (.25) -.000	B	+.010 C (.25) -.000	+.008 D (.20) -.000	E
01	1.665 (42.29)	1.376 (34.95)	1.096 (27.84)	.996 (25.30)	.688 (17.48)
02	2.246 (57.05)	1.960 (49.78)	1.686 (42.82)	1.581 (40.16)	.980 (24.89)

1/ Metric equivalent are given for general information only. Millimeters are in parentheses.

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Custodian:
Air Force - 17

Review activities.
Air Force - 11, 17, 85
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5935-F294-3)

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